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# MyDesign Portfolio

## *Element C Initial Template*

### How to use this template

Teachers: You may customize this document before providing it to students.

Students: The template below is one way to present your design requirements, but the specific assignment given by your instructor may require a different structure or dictate different details. If you make a copy of this template to edit for your own design requirements, be sure to delete the instructions. Make sure to list your design requirements in ranked order, with design requirement #1 having the highest priority. If possible, list 5 requirements that are labeled as HIGH priority and 1-5 additional requirements that are labeled as LOW priority.

### Element C: Presentation and Justification of Solution Design Requirements

#### Design Requirement #1 (HIGH Priority)

Write out this design requirement in a single sentence, ensuring that it is clear, detailed, and measurable.

The backpack hanger must stay securely attached to a desk or table for at least 6 continuous hours while supporting weight without slipping.

Explain how this design requirement directly ties to your prior research.

This requirement connects to our research because students are in school for about 6–7 hours per day, and backpacks need to stay off the floor the entire time to avoid clutter and safety issues.

Share this design requirement with (a) primary stakeholder(s) to get their validation. Share any feedback, notes, and/or your revisions as evidence that you conducted primary stakeholder validation.

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A student stakeholder said, "If it falls off during class, it defeats the whole purpose." Based on this feedback, long-lasting attachment was made a top priority.

### **Design Requirement #2 (HIGH Priority)**

Write out this design requirement in a single sentence, ensuring that it is clear, detailed, and measurable. The hanger must be able to support at least 20 pounds of weight without bending or breaking.

Explain how this design requirement directly ties to your prior research.

Our research showed that the average student backpack weighs around 15 pounds. Designing the hanger to hold 20 pounds provides a safety margin for heavier backpacks.

Share this design requirement with (a) primary stakeholder(s) to get their validation. Share any feedback, notes, and/or your revisions as evidence that you conducted primary stakeholder validation. A student stakeholder said, "Most backpacks aren't that heavy, but some definitely are," which confirmed the need for extra strength.

### **Design Requirement #3 (HIGH Priority)**

Write out this design requirement in a single sentence, ensuring that it is clear, detailed, and measurable. The hanger must be made from a rigid and durable material that can withstand daily use for at least one school year.

Explain how this design requirement directly ties to your prior research.

Research showed that classroom equipment goes through frequent wear and tear, so a weak material would break quickly and become unsafe.

Share this design requirement with (a) primary stakeholder(s) to get their validation. Share any feedback, notes, and/or your revisions as evidence that you conducted primary stakeholder validation.

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A student stakeholder said, “People might mess with it or kick it,” reinforcing the need for a strong material.

#### **Design Requirement #4 (HIGH Priority)**

Write out this design requirement in a single sentence, ensuring that it is clear, detailed, and measurable. The hanger must not distract students and should avoid bright colors or noisy, moving parts.

Explain how this design requirement directly ties to your prior research. Our research emphasized that distractions in the classroom can reduce focus and learning, so the design should blend in.

Share this design requirement with (a) primary stakeholder(s) to get their validation. Share any feedback, notes, and/or your revisions as evidence that you conducted primary stakeholder validation. A student stakeholder recommended using a dark color so it does not stand out or distract students.

#### **Design Requirement #5 (HIGH Priority)**

Write out this design requirement in a single sentence, ensuring that it is clear, detailed, and measurable.

The hanger must be foldable, compatible with most school desks, and no thicker than 1 inch when folded.

Explain how this design requirement directly ties to your prior research. This connects to our research because schools use many different desk styles, and a foldable design prevents it from getting in the way when not in use.

Share this design requirement with (a) primary stakeholder(s) to get their validation. Share any feedback, notes, and/or your revisions as evidence that you conducted primary stakeholder validation. A student stakeholder suggested a flexible and foldable design to improve usability.

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### **Design Requirement #6 (LOW Priority)**

Write out this design requirement in a single sentence, ensuring that it is clear, detailed, and measurable.

The hanger should be able to attach to the back of a chair if a desk attachment is not possible.

Explain how this design requirement directly ties to your prior research.

This ties to research showing that some desks do not have suitable edges, and chair attachment would keep backpacks off the floor.

Share this design requirement with (a) primary stakeholder(s) to get their validation. Share any feedback, notes, and/or your revisions as evidence that you conducted primary stakeholder validation.

A student stakeholder agreed this would be helpful in certain classrooms.

### **Design Requirement #7 (LOW Priority)**

Write out this design requirement in a single sentence, ensuring that it is clear, detailed, and measurable. The hanger should be black or another dark neutral color.

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Explain how this design requirement directly ties to your prior research. This connects to our research because dark colors match most classroom furniture and reduce visual distraction.

Share this design requirement with (a) primary stakeholder(s) to get their validation. Share any feedback, notes, and/or your revisions as evidence that you conducted primary stakeholder validation. A student stakeholder agreed that black would look clean and blend in..